CCTV Camera Setup

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Purpose:

The purpose of this lab is to configure camera coverage for my IT classroom. This is needed for testing site certifications and to promote a productive and ethical learning environment. Finally, it is meant to ensure accountability when and if valuable equipment is lost.

Definitions:

CCTV:

Closed-Circuit Television; A type of surveillance camera that is designed to only provide coverage and accessibility to a predefined set of monitoring stations.

DVR:

Digital Video Recorder; A useful technology to integrate with CCTV to allow for storage of video to enable checking past footage.

UPS:

Uninterruptable Power Supply; A device that ensures consistent power to critical devices during a loss of power, these often run off battery power but

can also be outfitted with generators to provide longer term power solutions.

HAZMAT:

Hazardous Material; A general term referring to any matter that is toxic, radioactive, oxidative, flammable, or biologically dangerous.

PSU:

Power Supply Unit; A separate electrical component designed to give a reliable source of power to simplify overall component design

Background Information:

CCTV:

CCTV surveillance was invented in 1927 by Russian physicist Leon Theremin and was used by Josef Stalin to monitor visitors entering and exiting the Kremlin. Over the last 98 years not much has changed. Nowadays it is used widely as a tool in monitoring everything from workplaces to subways. It is particularly useful in three key scenarios: First, to provide evidence in case of a crime/incident; Second, as a tool to provide cheap, 24/7 monitoring and surveillance; Third, when having someone present is not an option, such as in the case of a HAZMAT environment.

In our setup, we will deploy CCTV cameras to monitor activities in the classroom to promote an environment of learning and integrity that cannot be reliably reached in another way. The accountability that CCTV enforces also makes it required to certify our classroom as an exam site for various IT certifications.

DVRs:

Digital Video Recording is a cornerstone technology in most modern CCTV setups, mainly because it means that the CCTV footage does not need constant human monitoring and it also allows for playback of footage to provide as

evidence in the resolution of a crime/incident. This makes it incredibly important as it drastically reduces both costs and increases the utility of CCTV deployments.

In our deployment we save all camera footage to a 5.4 terabyte server to provide a lengthy rollback time ensuring plenty of time to find and create a backup should a situation mandate it.

Why Wired Connections are Better:

In our deployment, we use wired CCTV cameras for multiple reasons. First, and most importantly given the nature of this being an IT classroom it can be expected that there will be attempted tampering with and Wi-Fi based cameras, the nature of wireless networks makes them more susceptible to attacks and means that they are not viable as our chief monitoring hardware. A secondary reason is reliability; a wired connection means that fewer electronics need to be connected to a UPS to ensure that a power outage doesn't affect the efficacy of the camera deployment, additionally wired connections often have better operational reliability than Wi-Fi connections where there are more points of failure and a generally more unstable connection.

Summary:

In this lab I set up 14 wired CCTV cameras to provide coverage of two connected classrooms. Additionally, a monitoring and recording station was added to centralize the management and use of the cameras to a location readily accessible to the administrators.

Setup:

I started by first analyzing potential camera locations and quickly found that I would need to purchase conduit mounts due to the lack of usable attachment surfaces. Once I found a suitable mount, I moved on to cataloging what we currently had and what we would need, I found that we already had the needed cameras and most of the needed wires. However, after estimating the needed cable lengths for various camera locations, I found that on one of them I would need to find a 150 wire. This was easy enough and with it delivered I began to install the cameras. I began to first put all of them in their respective places without any of the wires, when this was finished I started at the furthest camera

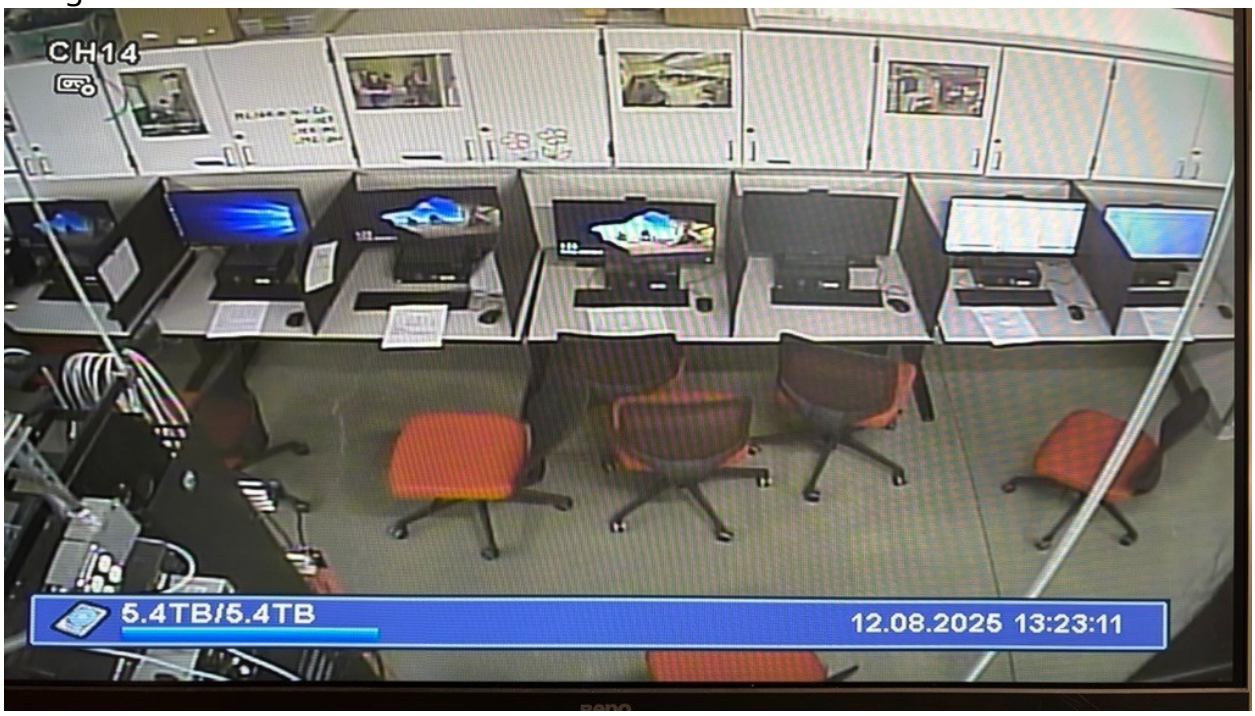
and moved towards the monitoring station, beginning a new wire whenever I got to the next camera, this went on until I had hooked up all of the cameras and with a monitor and power supply added, I had a working setup.

Pictures:

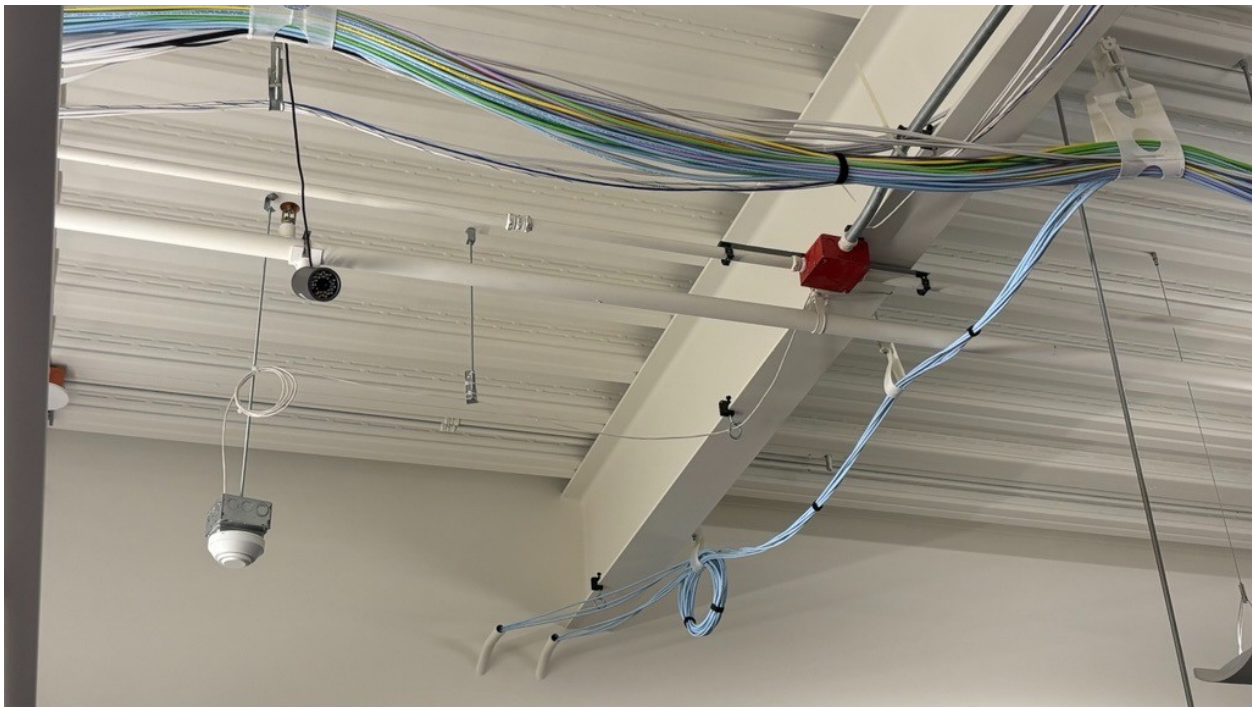
Here Pictured is the DVR/Camera Controller with the PSU stacked on top.

Here is a view of the screen in a 16-camera view, notice the two separate rooms.

A close-up view of one of the camera views looking at a row of computers used in testing.



Here is a picture of one of the installed cameras.



Problems:

My first problem was installing some of the camera's upside down, I had been unsure of which direction they went and inevitably a few of them were upside down, thankfully the fix was as simple as unscrewing them and reinstalling the camera flipped right side up.

My next issue was having extra length on the hose clamps holding the mounts to the conduits and sagging wires blocking the camera view. With some tin snips, the clamps were shortened and pulling the wires tighter and rerouting fixed the wire sag issues.

Finally, one of the installed cameras had a fogged lens, which meant that the camera had to be replaced with an extra; the new camera worked great and resolved the issue.

While not exactly an issue, I also had to swap around which port cameras were plugged into until I had a logical setup that could be flipped through sequentially in a manner that was easy to memorize and control.